



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Mohammed Razia Alangir Banu
Student Name	N. Siri Chandana Reddy
Roll Number	2520030142
Branch	CSE
Course Outcome	CO4 (Dijkstra's Algorithm)

Case Study Topic: Waste Sense – Smart Waste Collection & Recycling Optimization System Using Dijkstra's

Work Done Summary:

The Waste Sense system uses Dijkstra's Algorithm to determine the shortest and most cost-effective routes for waste collection vehicles. The algorithm helps reduce transportation costs, fuel consumption, and collection time by identifying the minimum-distance path between collection centers, smart bins, and recycling plants.

Implementation Details:

1. Create a weighted graph for the waste collection network.
2. Add smart bins, collection centres, and recycling plants as nodes.
3. Add roads with transportation costs as weighted edges.
4. Select a source collection centre.
5. Apply Dijkstra's Algorithm to calculate the shortest distance to all locations.
6. Display the optimal waste collection routes and their costs.

Result: Screenshot/output:

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Locations:
Collection Center
Bin A
Bin B
Bin C
Recycling Plant

BFS Traversal:
Collection Center -> Bin A -> Bin B -> Bin C -> Recycling Plant

DFS Traversal:
Collection Center -> Bin A -> Bin C -> Recycling Plant -> Bin B
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Dijkstra's Algorithm provides an effective solution for route optimization in smart waste management systems. By selecting the shortest paths, the Waste Sense system ensures efficient waste collection, reduced operational expenses, and better utilization of transportation resources.

GIT HUB LINK : <https://github.com/nemaliscreddy78-glitch/DSA2>

Student Signature

Faculty Signature